

Fault register

Every fix the canonical single-sheet schematic needs before VAL-G2 can close. 65 repairs, 89 declared transactions, ordered so that no transaction invalidates the evidence of one before it. Every entry is a proposal — the single writer reconfirms against live before acting.

65 REPAIRS	89 TRANSACTIONS	22 P0	19 P1	18 P2
6 P3	18 NEED A RULING	6 FULLY STALLED		

CAP-01 — No transaction in this queue can CLOSE until readable capture works.

harness/easyeda_readable_capture.mjs refuses at a measured 0.2 px per schematic unit against a floor of 1. Canon section 8 requires a block-scale image that makes the changed objects readable before a transaction may be inspected and closed. At 0.2 px/unit a 0402 value string, a wire endpoint and a No-Connect cross are all sub-pixel. Executing without closing leaves the gate IN_FLIGHT, which canon K1E-041 names as more dangerous than a clean failure.

Do these first

Read-only. Neither is a canvas mutation, and both change what the repairs below are allowed to assume.

P-1

READ-ONLY CENSUS

Live No-Connect census

NEW FINDING raised by this consolidation, present in no lane. Between the 12:17 and 14:58 DRC runs, J1-PWR1.A6/.A7/.B6/.B7 (the four USB data contacts), J1-PWR1.A8/.B8 (SBU) and D1-PWR1.3/.4/.5/.6

all left the floating-pin list. EasyEDA omits a pin from that list when it is wired OR when it carries a No-Connect flag. Nothing in the ledger records those pins being wired; the ledger does record 'Captain applied unused-pin NC crosses' in that window - the same sweep the motion lane identified as resolving 95 pins across 17 unrelated blocks.

INFERENCE [INFERENCE, high confidence] those ten pins now carry No-Connect flags, so the sheet formally certifies as deliberately unused the exact pins USB-TOPOLOGY-AUDIT.md S2 and PIN-AUDIT-PWR1.md S(D1) call P0 defects. Same mechanism as NFC-D16 for U12-NFC.15 (RFO2), and the same class as the INA226 address flags.

ACTION Enumerate every NO_CONNECT attribute on the live document with its parent pin, and diff against the 12 present in the frozen source. Publish the list.

SETTLES RQ-004 RQ-005 RQ-062 RQ-063 disagreement-5

RQ-056 EVIDENCE EXPORT

Export this board's BOM and CPL and read the Comment column

SETTLES Whether Add into BOM behaves on K1 as measured on IM69D130 (5 designators in the CPL but not the BOM, 0 the other way) Whether Comment is fed by Name or by Value - if by Value, every K1 BOM line reads RC0402FR-0710KL instead of 10k, and BOM-SEMANTIC-AUDIT S6.3's INFO becomes a real finding disagreement-4

The fixes

In execution order. The stripe is severity; the outlined chip is how strong the evidence is — **netlist-confirmed** means EasyEDA's own DRC names it, **datasheet-derived** means a vendor document says the circuit is wrong and no automated check on this host will ever see it.

All severities

P0

P1

P2

P3

All evidence ▼

All blocks ▼

89 of 89 shown

1	RQ-001DELETENETLIST-CONFIRMEDP2nfc-regulator-decouple Debris, almost certainly an aborted or misplaced wire during canonical-nfc-regulator-decouple-*. Must precede RQ-028, which adds parts to the same row. OBJECTS wire e153914 NETS <none - the wire carries NO NET attribute>
---	--

EXPECTED RESULT	wire count -1; single-pin network \$1N153914 disappears; unnamed_wires returns to 0
VISUAL DELTA	one 85-unit horizontal segment vanishes from the gap between two populated capacitor rows
HOW TO CHECK	re-run DRC: \$1N153914 absent from the single-network list; no capacitor in the row loses a connection; C92-C97 pin membership unchanged

2

RQ-002 DELETE NETLIST-CONFIRMED P0 power-buck

Root cause is the Y-negation trap applied in reverse: a repair script read a pin coordinate (negative y) and wrote a wire at that literal value instead of negating it.

OBJECTS	wire e146347
NETS	BUCK_PG
EXPECTED RESULT	BUCK_PG loses its off-sheet member; wire count -1
VISUAL DELTA	nothing changes on the visible page - which is the point, and why this must be verified by source read-back rather than by eye
HOW TO CHECK	source read-back: zero wires with any negative coordinate; BUCK_PG reaches exactly one pin (R75-PWR2.1) immediately after this transaction

3

RQ-003 WIRE NETLIST-CONFIRMED P0 power-buck after RQ-002

BUCK_PG becomes a two-pin net (R75-PWR2.1 + U3-PWR2.5); U3-PWR2.5 leaves the floating list; BUCK_PG leaves the single-pin-network list

OBJECTS	new BUCK_PG wire R75-PWR2.1 U3-PWR2.5 (PG)
NETS	BUCK_PG
EXPECTED RESULT	BUCK_PG becomes a two-pin net (R75-PWR2.1 + U3-PWR2.5); U3-PWR2.5 leaves the floating list; BUCK_PG leaves the single-pin-network list
VISUAL DELTA	a visible wire from the 10 k pull-up to the TPS62913 PG pin
HOW TO CHECK	DRC: BUCK_PG no longer a single network AND U3-PWR2.5 no longer floating. Only when this closes does D-045 become IMPLEMENTED; until then the register must read ratified-not-implemented.

4

RQ-004 REPAIR NETLIST-CONFIRMED P0 usb-inlet after P-1

four pins return to the floating list; the sheet stops certifying D+/D- as deliberately unused

OBJECTS	No-Connect flags on J1-PWR1.A6, .B6, .A7, .B7
NETS	
EXPECTED RESULT	four pins return to the floating list; the sheet stops certifying D+/D- as deliberately unused

VISUAL DELTA	four NC crosses disappear from the J1 symbol
HOW TO CHECK	live NC census shows zero flags on A6/A7/B6/B7; DRC floating count for J1-PWR1 rises from 9 to 13 - a RISE is the pass condition here. Keep .A8/.B8 (SBU) flagged; those are correct.

5

RQ-005 REPAIR NETLIST-CONFIRMED P0 usb-protection after P-1

Pin 5 is the VBUS clamp node. With it NC-flagged, the sheet certifies that a TVS allowed to clamp nothing is correct.

OBJECTS	No-Connect flags on D1-PWR1.3, .4, .5, .6
NETS	
EXPECTED RESULT	D1 returns four pins to the floating list
VISUAL DELTA	four NC crosses disappear from the USBLC6-2SC6 symbol
HOW TO CHECK	live NC census clean on D1; DRC lists D1-PWR1.3/4/5/6 floating again

6

RQ-006 REPAIR NETLIST-CONFIRMED P1 power-monitor

Present in the frozen source - 2 of its 12 NC marks. A No-Connect flag is an assertion that the open state is correct; here it silences a DRC warning about a pin the vendor requires to be driven, so the annotation certifies the defect.

OBJECTS	No-Connect flags on U2-PWR1.1 (A1) and U2-PWR1.2 (A0)
NETS	
EXPECTED RESULT	INA226 address pins return to floating pending RQ-007
VISUAL DELTA	two NC crosses disappear
HOW TO CHECK	source read-back: no_connect_parent_ids no longer contains the U2-PWR1 pins

7

RQ-007 WIRE DATASHEET-DERIVED P1 power-monitor after RQ-006

both pins join GND (address 1000000) or the pair chosen from TI SBOS547C Table 6-2; the I2C address becomes defined

OBJECTS	U2-PWR1.1 (A1) U2-PWR1.2 (A0)
NETS	GND
EXPECTED RESULT	both pins join GND (address 1000000) or the pair chosen from TI SBOS547C Table 6-2; the I2C address becomes defined
VISUAL DELTA	two short wires from the INA226 address pins to the GND rail
HOW TO CHECK	DRC: neither pin floating, neither NC-flagged. The board can now enumerate its own current monitor.

8

RQ-008

WIRE

NETLIST-CONFIRMED

P0

motion

three pins leave the floating list; the LIS2DH12 gets its full ground reference

OBJECTS

U13-MOT.5 (RES)

U13-MOT.7 (GND)

U13-MOT.8 (GND)

NETS

GND

EXPECTED RESULT

three pins leave the floating list; the LIS2DH12 gets its full ground reference

VISUAL DELTA

three wires from the LGA-12 to the GND rail

HOW TO CHECK

DRC: U13-MOT pins 5, 7, 8 absent from the floating list

9

RQ-009

WIRE

NETLIST-CONFIRMED

P0

motion

blocked · DEC-11

CS is the highest-severity pin in the motion block.

OBJECTS

U13-MOT.2 (CS)

U13-MOT.3 (SD0/SA0)

NETS

3V3 GND or 3V3 per DEC-11

EXPECTED RESULT

two pins leave the floating list; I2C mode is forced and the 7-bit address becomes deterministic

VISUAL DELTA

two wires from the sensor to the 3V3 rail / GND

HOW TO CHECK

DRC: pins 2 and 3 absent from the floating list; the resulting 7-bit address recorded in the D18 hardware-to-firmware contract

10

RQ-010

REPAIR

NETLIST-CONFIRMED

P0

nfc-clock

This finding has survived three independent measurements: the geometry oracle, the repo harness, and two GUI DRC runs 2h41m apart. DRC caught the two floating pads but NOT the fact that NFC_XTO had landed on a ground pad - only half of it.

OBJECTS

Y2-NFC.2

Y2-NFC.3

Y2-NFC.4

C54-NFC

C55-NFC

NETS

NFC_XTO

GND

EXPECTED RESULT

the 27.12 MHz oscillator loop closes; Y2-NFC.3 and .4 leave the floating list

VISUAL DELTA

the crystal's wiring visibly changes terminal; two case pads gain ground ties

HOW TO CHECK

DRC: Y2-NFC.3 and Y2-NFC.4 absent from the floating list. Verify the symbol pin numbering against the Abracon ABM12 datasheet FIRST - NFC_XTO currently lands on pin 2, which the symbol names GND. Re-derive C54/C55 from the crystal's specified load capacitance rather than keeping 10 pF.

11

RQ-011

WIRE

NETLIST-CONFIRMED

P0

supervisor

blocked · DEC-05

Pair with RQ-059. Wiring SENSE without a consumer, or a consumer without SENSE, still leaves RT_PWR_VALID decorative. Do not conflate BUCK_PG (RQ-003) with the supervisor's

RESET# - different signals, different meanings, neither substitutes for the other.

OBJECTS	U16-VAL.5 (SENSE)
NETS	3V3 or 1V15_CORE per DEC-05
EXPECTED RESULT	the TPS3808 begins monitoring something; RT_PWR_VALID stops being decorative at the source end
VISUAL DELTA	one wire from the supervisor's SENSE pin to the chosen rail
HOW TO CHECK	DRC: U16-VAL.5 absent from the floating list

12

RQ-012 WIRE NETLIST-CONFIRMED P0 usb-inlet

Couples to RQ-021: raising the eFuse ILIM only helps once all four VBUS contacts are bonded, or the connector becomes the binding limit instead of the eFuse.

OBJECTS	J1-PWR1.A4 J1-PWR1.A9 J1-PWR1.B9
NETS	5V_USB
EXPECTED RESULT	all four VBUS contacts bonded; the connector's collective 5.00 A rating becomes available instead of one contact's 1.25 A
VISUAL DELTA	three wires from the receptacle's VBUS contacts to the inlet net
HOW TO CHECK	DRC: A4/A9/B9 absent from the floating list

13

RQ-013 WIRE NETLIST-CONFIRMED P0 usb-inlet

all four GND contacts and the shell bonded

OBJECTS	J1-PWR1.A1 J1-PWR1.A12 J1-PWR1.B12 J1-PWR1.1 (EH shell)
NETS	GND
EXPECTED RESULT	all four GND contacts and the shell bonded
VISUAL DELTA	four wires to the ground rail
HOW TO CHECK	DRC: A1/A12/B12 and pin 1 absent from the floating list

14

RQ-014 REPAIR DATASHEET-DERIVED P0 validation-header

Captain's re-derivation is CONFIRMED in part (it is a single strap, not a differential pair - OPT_USB_AUD has exactly two endpoints, OPT_USB_AUD_RT exactly two, and there is no D+ counterpart anywhere) and REFUTED in part (the RT-side endpoint is not an option GPIO). As drawn with R56 unfitted it is inert; fitted, it would bias the USB D- line to the header's logic level.

OBJECTS	U6-RTC.M8 R56-VAL (DNP) J11-VAL.2
NETS	OPT_USB_AUD OPT_USB_AUD_RT

EXPECTED RESULT	M8 leaves OPT_USB_AUD_RT and becomes free for USB_OTG1_DN; the validation strap lands on a pin that can serve it
VISUAL DELTA	the strap visibly moves off the RT1062's USB ball
HOW TO CHECK	source read-back: U6-RTC.M8 has no net until RQ-016 assigns it

15 **RQ-015-P** PLACE DATASHEET-DERIVED P0 usb-inlet
component count +2

OBJECTS	two new 5.1 k0hm 0402 resistors, Rd for CC1 and CC2
NETS	
EXPECTED RESULT	component count +2
VISUAL DELTA	two new 0402 bodies beside the J1 receptacle
HOW TO CHECK	both bodies placed, non-overlapping, inside the sheet frame, in the J1 block

16 **RQ-015-D** DESIGNATE DATASHEET-DERIVED P0 usb-inlet after RQ-015-P
two new designators in the PWR1 domain with value 5.1k and a real LCSC code

OBJECTS	the two new Rd resistors
NETS	
EXPECTED RESULT	two new designators in the PWR1 domain with value 5.1k and a real LCSC code
VISUAL DELTA	designator and value text appears on both bodies
HOW TO CHECK	designators unique; value 5.1k; Supplier Part carries an LCSC Cxxxxxxx code, not an MPN.n device key

17 **RQ-015-W** WIRE DATASHEET-DERIVED P0 usb-inlet after RQ-015-D

See DEC-02: if option (a) is chosen, both CC nodes must also be tapped to RT1062 ADC inputs through protection dividers in the same edit.

OBJECTS	J1-PWR1.A5 (CC1) J1-PWR1.B5 (CC2) the two Rd resistors
NETS	USB_CC1 USB_CC2 GND
EXPECTED RESULT	J1 becomes a Type-C sink; CC1/CC2 leave the floating list
VISUAL DELTA	two wires from the CC contacts through the resistors to GND
HOW TO CHECK	DRC: J1-PWR1.A5 and .B5 absent from the floating list; each CC pin independently terminated, as J7 already does with R21/R22-ESP

18 **RQ-016-P** PLACE DATASHEET-DERIVED P0 usb-data after RQ-005
component count +2; D1 repositioned

OBJECTS	D1-PWR1 (USBL C6-2SC6, relocated into the J1 data path) two series TUNE footprints
NETS	
EXPECTED RESULT	component count +2; D1 repositioned
VISUAL DELTA	the ESD array moves into the data path; two new footprints appear between it and the RT1062
HOW TO CHECK	D1 sits between J1 and the RT1062 in the drawing, not on the 5V_PROTECTED rail

19

RQ-016-D DESIGNATE DATASHEET-DERIVED P0 usb-data after RQ-016-P

two new designators, value 0R, FITTED

OBJECTS	the two TUNE footprints
NETS	
EXPECTED RESULT	two new designators, value 0R, FITTED
VISUAL DELTA	designator and value text appears
HOW TO CHECK	value is 0 Ohm fitted by default, NOT a populated non-zero resistor

20

RQ-016-W WIRE DATASHEET-DERIVED P0 usb-data after RQ-016-D, RQ-004, RQ-014

This is D-044's unimplemented D+/D- clause. The USBL C6-2SC6 is the correct class of part for this job (2.5 pF typ / 3.5 pF max line-to-ground) and this is exactly the application it is designed for - unlike its present misuse as a rail TVS.

OBJECTS	J1-PWR1.A6+B6 tied J1-PWR1.A7+B7 tied D1-PWR1 pins 1,6,3,4,5,2 U6-RTC.L8 (USB_OTG1_DP) U6-RTC.M8 (USB_OTG1_DN)
NETS	USB_DP USB_DM USB_DP_ESD USB_DM_ESD GND protected VBUS
EXPECTED RESULT	J1 D+/D- reach the RT1062; D1 becomes electrically live for the first time; L8 leaves the floating list
VISUAL DELTA	four wires from the receptacle through the ESD array and TUNE footprints to the RT1062
HOW TO CHECK	DRC: J1.A6/A7/B6/B7 and U6-RTC.L8 all absent from the floating list. D+ in on D1 pin 1 and out on pin 6; D- in on pin 3 and out on pin 4; pin 5 to the VBUS being protected; pin 2 to GND - the same arrangement U10-ESP already uses correctly on J7.

21

RQ-017-P PLACE DATASHEET-DERIVED P1 usb-data blocked · DEC-02

component count +1

OBJECTS	one new 1 uF / 10 V 0402 capacitor
NETS	

EXPECTED RESULT	component count +1
VISUAL DELTA	one new body at the RT1062's USB VBUS ball
HOW TO CHECK	placed adjacent to U6-RTC.N6

22

RQ-017-D DESIGNATE DATASHEET-DERIVED P1 usb-data after RQ-017-P

one new designator, value 1uF, 10 V rating recorded

OBJECTS the new VBUS capacitor

NETS

EXPECTED RESULT one new designator, value 1uF, 10 V rating recorded

VISUAL DELTA designator and value text

HOW TO CHECK value 1 uF, voltage rating 10 V per NXP HDG Table 2

23

RQ-017-W WIRE DATASHEET-DERIVED P1 usb-data blocked · DEC-02 after RQ-017-D

The margin is ZERO: absolute maximum 5.50 V, and USB VBUS is specified to 5.50 V with hot-plug overshoot on top. That is why DEC-02 exists. 5V_PROTECTED is preferred because it reuses protection that must exist anyway - at the cost that the RT sees 'VBUS present' only after the eFuse has enabled, which is the correct semantic for a self-powered device.

OBJECTS U6-RTC.N6 (USB_OTG1_VBUS) the new 1 uF capacitor

NETS 5V_PROTECTED (per DEC-02) GND

EXPECTED RESULT N6 leaves the floating list; session-valid detection becomes possible

VISUAL DELTA one wire plus one decoupling leg at the RT1062

HOW TO CHECK DRC: N6 absent from the floating list. NO divider - NXP specifies none, and a divider would defeat the session-valid detection the pin exists to perform.

24

RQ-018-P PLACE DATASHEET-DERIVED P0 power-entry

component count +1

OBJECTS one unidirectional VBUS TVS

NETS

EXPECTED RESULT component count +1

VISUAL DELTA one new body between the connector and F1-PWR1

HOW TO CHECK placed on the 5V_USB side, UPSTREAM of F1 and U1

25

RQ-018-D DESIGNATE DATASHEET-DERIVED P0 power-entry after RQ-018-P

one new designator with a real MPN and LCSC code

OBJECTS	the new inlet TVS
NETS	
EXPECTED RESULT	one new designator with a real MPN and LCSC code
VISUAL DELTA	designator and value text
HOW TO CHECK	working voltage above 5.5 V; clamping voltage comfortably below the eFuse's 28 V absolute maximum; sized for inlet surge, not for ESD

26

RQ-018-W WIRE DATASHEET-DERIVED P0 power-entry after RQ-018-D

The part that needs protecting from an inlet transient is the eFuse itself - TPS259474L absolute maximum 28 V, and it is the first silicon the transient reaches. Protection placed behind it protects nothing that was at risk. ST's USBLC6-2SC6 gives NO continuous current rating anywhere and is a 2.5 pF low-capacitance array built for D+/D-, not a bulk VBUS TVS.

OBJECTS	the inlet TVS
NETS	5V_USB GND
EXPECTED RESULT	the inlet gains real transient protection for the first time
VISUAL DELTA	two wires from the TVS to the inlet net and ground
HOW TO CHECK	source read-back: the TVS sits on 5V_USB, not on 5V_PROTECTED

27

RQ-019 REPAIR DATASHEET-DERIVED P0 power-entry blocked · DEC-08

Two independent problems wanting different fixes: rating AND position. FB1/FB2 (LED branches, ≤ 1.05 A) and FB3 (NFC 5 V, ≤ 0.5 A peak) use the same part CORRECTLY - only F1 sees the whole trunk. Position: F1 upstream of the eFuse followed by C1 22 uF forms an LC tank across hot-plug, and the overshoot lands on an eFuse whose OVLO is 5.46 V (RQ-020).

OBJECTS	F1-PWR1 (BLM21PG221SN1D)
NETS	5V_USB 5V_USB_FILTERED
EXPECTED RESULT	the trunk series element stops being the binding limit, or is removed from the trunk entirely
VISUAL DELTA	one part swaps, or one part disappears and the trunk becomes continuous
HOW TO CHECK	BOM shows a bead rated ≥ 3 A with DCR ≤ 20 mOhm, or F1 is absent from the trunk

28

RQ-020 REPAIR DATASHEET-DERIVED P0 power-entry

A compliant source at the high end of its range trips the board off before any fault exists - and with F1's LC tank ahead of it, hot-plug ringing reaches that threshold from a source nowhere near 5.5 V steady-state.

OBJECTS	R63-PWR1 (1.05 M) R2-PWR1 (100 k) R64-PWR1 (324 k) U1-PWR1.2 (OVL0)
NETS	
EXPECTED RESULT	the eFuse stops latching off inside the USB VBUS tolerance band
VISUAL DELTA	two or three resistor value texts change
HOW TO CHECK	recompute $V_{IN(OV)} = 1.20 \times (R63+R2+R64)/R64 \geq 6.0$ V, still far below the 23 V operating and 28 V absolute maximum

29

RQ-021 REPAIR DATASHEET-DERIVED P1 power-entry

Do this together with RQ-046 (rebinding) - the repair is not 'correct the binding to match the drawing', because the drawn value was chosen against the inherited figures the envelope replaces. Couples to DEC-02 and RQ-012: raising ILIM makes the CONNECTOR the binding limit unless all four J1 VBUS contacts are bonded. The window carries only 2.4% of total slack ($3.000/2.397 = 1.252$ against the part's own 1.222 tolerance span).

OBJECTS	R1-PWR1
NETS	USB_EFUSE_ILIM
EXPECTED RESULT	trunk eFuse limit moves from 2.507 A typ / 2.26 A min to 2.689 A typ / 2.42 A min / 2.958 A max
VISUAL DELTA	one value text changes from 1.33k to 1.24k
HOW TO CHECK	$R_{ILM} = 3334 / I_{LIM}$ (TI SLVSFC9C Eq. 5). Permitted window 1223-1252 Ohm; 1.24 kOhm (E96) is the ONLY standard value inside it. 1.20k max exceeds the 3.0 A source; 1.30k and the drawn 1.33k both put the guaranteed minimum below the 2.40 A design point.

30

RQ-022-P PLACE DATASHEET-DERIVED P2 power-monitor

component count +3

OBJECTS	two ≤ 10 Ohm series resistors one 0.1-1 uF capacitor
NETS	
EXPECTED RESULT	component count +3
VISUAL DELTA	three new bodies at the shunt
HOW TO CHECK	placed between RSH1-PWR1 and U2-PWR1 pins 9/10

31

RQ-022-D DESIGNATE DATASHEET-DERIVED P2 power-monitor after RQ-022-P

 three new designators with values ≤ 10 Ohm and 0.1-1 uF

OBJECTS	the three new filter parts
NETS	

EXPECTED RESULT	three new designators with values ≤ 10 Ohm and 0.1-1 uF
VISUAL DELTA	designator and value text on all three
HOW TO CHECK	series resistance 10 Ohm or less per TI SBOS547C S6.4.2

32

RQ-022-W WIRE DATASHEET-DERIVED P2 power-monitor after RQ-022-D

Enforcing Kelvin is a VAL-G3 layout constraint. What is in scope here is making layout INHERIT a constraint rather than a hope. Sense topology is otherwise correct and verified: IN+ on 5V_PROTECTED, IN- on 5V_SYS, RSH1 bridging exactly those two nets; 20.8 mV / 43 mW at the 2.08 A peak against an 81.92 mV full scale.

OBJECTS U2-PWR1.9 (IN-) U2-PWR1.10 (IN+) RSH1-PWR1

NETS RSH1_KELVIN_P RSH1_KELVIN_N

EXPECTED RESULT the measurement node stops inheriting eFuse hot-swap slew and LED switching; dedicated Kelvin nets exist for layout to inherit

VISUAL DELTA the INA226 inputs visibly reach the shunt through the new resistors

HOW TO CHECK source read-back: IN+/IN- reach the shunt on dedicated Kelvin nets, not directly on 5V_PROTECTED / 5V_SYS

33

RQ-023-P PLACE DATASHEET-DERIVED P1 mic-rail

component count +1

OBJECTS one ≥ 1 uF ceramic capacitor

NETS

EXPECTED RESULT component count +1

VISUAL DELTA one new 0402 at the TLV75533 output

HOW TO CHECK placed adjacent to U5-PWR2 pin 5

34

RQ-023-D DESIGNATE DATASHEET-DERIVED P1 mic-rail after RQ-023-Pone new designator, value ≥ 1 uF

OBJECTS the new LDO output capacitor

NETS

EXPECTED RESULT one new designator, value ≥ 1 uF

VISUAL DELTA designator and value text

HOW TO CHECK value ≥ 1 uF ceramic

35

RQ-023-W WIRE DATASHEET-DERIVED P1 mic-rail after RQ-023-D

Measured: 3V3_MIC_REG has exactly two wires and two pins with NO dangling stub, so no hidden capacitor exists. C15-PWR2 10 uF sits on Q1's DRAIN, downstream of the load switch - it cannot stabilise the regulator loop.

OBJECTS	U5-PWR2.5 (OUT) the new capacitor
NETS	3V3_MIC_REG GND
EXPECTED RESULT	3V3_MIC_REG reaches three pins instead of two; the TLV75533 gets an output capacitor for the first time
VISUAL DELTA	one decoupling leg appears at the LDO output
HOW TO CHECK	source read-back: 3V3_MIC_REG membership grows from {U5-PWR2.5, Q1-PWR2.2} to include the new capacitor

36

RQ-024 REPAIR DATASHEET-DERIVED P2 power-buck

V_OUT is presently $0.8 \times (1 + 100k/32.4k) = 3.269 \text{ V}$, -0.94% from 3.3 V - within every downstream tolerance and NOT to be chased on its own. The BOM binding of both parts verifies clean (PIN-AUDIT-PWR1 positive verification); only the value is wrong.

OBJECTS	R6-PWR2 (32.4 k) R5-PWR2 (100 k)
NETS	
EXPECTED RESULT	the low-noise property that is the only reason to select a TPS62913 is restored
VISUAL DELTA	two value texts change
HOW TO CHECK	$R6 \leq 5 \text{ k}\Omega$ with R5 rescaled to hold V_OUT near 3.3 V

37

RQ-025 WIRE DATASHEET-DERIVED P2 power-entry

Three unobservable power-good signals on a board whose stated mission is to maximise observability is a coherent gap, not three separate slips: U1's PG reaches R67 and nothing else, U4's PG is NC-flagged outright, and U3's PG is RQ-003.

OBJECTS	U1-PWR1.3 (PG/AUXOFF) an RT1062 GPIO
NETS	USB_EFUSE_PG
EXPECTED RESULT	at least one power-good signal becomes observable by firmware
VISUAL DELTA	one new wire leaving the power block toward the RT1062
HOW TO CHECK	source read-back: USB_EFUSE_PG reaches a processor pin, not just R67-PWR1

38

RQ-026 REPAIR DATASHEET-DERIVED P0 nfc-supply blocked • DEC-03

Part-damaging, not a performance issue. Invisible to DRC - both pins ARE connected, just to different rails. Also unsettles the 0.500 A NFC term in the trunk envelope until resolved.

Changes architecture/POWER-ARCHITECTURE.md. NFC_5V keeps its ferrite and bulk whichever rail is chosen.

OBJECTS	U12-NFC.8 (VDD) U12-NFC.10 (VDD_TX)
NETS	3V3 NFC_5V
EXPECTED RESULT	the 1.7 V supply split collapses to 0 V; both pins sit on one rail
VISUAL DELTA	one supply wire moves rail
HOW TO CHECK	source read-back: U12-NFC.8 and .10 share a net. VDD_IO is genuinely independent and may stay at 3.3 V.

39

RQ-027-P PLACE DATASHEET-DERIVED P0 nfc-rf blocked · DEC-04

component count +2 -1

OBJECTS	CVDR1 (series) CVDR2 (shunt) delete L3-NFC (33 nH)
NETS	
EXPECTED RESULT	component count +2 -1
VISUAL DELTA	two new caps at RFI1; one series inductor removed from the matching path
HOW TO CHECK	CVDR1 in series from the antenna node to RFI1; CVDR2 shunt from RFI1 to ground

40

RQ-027-D DESIGNATE DATASHEET-DERIVED P0 nfc-rf blocked · DEC-04

after RQ-027-P

seven values restamped as TUNE_TBD with the constraint recorded, not a number invented

OBJECTS	CVDR1 CVDR2 L2-NFC C59-NFC C60-NFC C61-NFC R42-NFC
NETS	
EXPECTED RESULT	seven values restamped as TUNE_TBD with the constraint recorded, not a number invented
VISUAL DELTA	value texts change across the RF stage
HOW TO CHECK	EMC cutoff constraint recorded as 8-17 MHz excluding 13-14 MHz; all matching values stamped provisional; AN5592 gives CVDR1 = 10 pF in all three worked topologies with CVDR2 at 15 pF for single-ended-with-cable

41

RQ-027-W WIRE DATASHEET-DERIVED P0 nfc-rf blocked · DEC-04 after RQ-027-D

ONE SOLVE, NOT THREE EDITS. L2 5.6 nH with C59 22 pF resonates at ~453 MHz - 27x the upper bound - and both values are wrong together, which reads as a copied set rather than a derived one. The divider appears INSIDE the matching equation ($CP = \dots - \frac{CVDR1 \cdot CVDR2}{(CVDR1 + CVDR2)}$), so it cannot be bolted on afterwards. Re-derive from STSW-ST25R004 against MEASURED antenna parameters.

OBJECTS	U12-NFC.22 (RFI1) NFC_ANT node CVDR1 CVDR2
NETS	NFC_ANT GND a new RFI1 divider node
EXPECTED RESULT	the receive path gains its attenuation; the EMC filter lands in band; the matching network matches an ST topology
VISUAL DELTA	RFI1 no longer sits on the raw antenna node
HOW TO CHECK	source read-back: RFI1 reaches the antenna only through CVDR1, with CVDR2 to ground. Voltage at the receive pins must not exceed 3 Vpp.

42

RQ-028-P PLACE DATASHEET-DERIVED P1 nfc-regulator-decouple after RQ-001

component count +5

OBJECTS	five new 10 nF capacitors
NETS	
EXPECTED RESULT	component count +5
VISUAL DELTA	five new 0402 bodies in the NFC regulator cap row
HOW TO CHECK	one per rail beside C92-C96; the row must already be clear of the RQ-001 debris wire

43

RQ-028-D DESIGNATE DATASHEET-DERIVED P1 nfc-regulator-decouple
after RQ-028-P

five new designators at 10 nF; C97 value changes from 2.2 uF to 1 uF

OBJECTS	the five new capacitors C97-NFC
NETS	
EXPECTED RESULT	five new designators at 10 nF; C97 value changes from 2.2 uF to 1 uF
VISUAL DELTA	designator and value text on five bodies; one value text changes
HOW TO CHECK	five at 10 nF; C97 at 1 uF with its own 10 nF partner

44

RQ-028-W WIRE DATASHEET-DERIVED P1 nfc-regulator-decouple after RQ-028-D

each regulator rail carries its specified parallel pair

OBJECTS	C92-NFC through C97-NFC and the five new partners
NETS	NFC_VDD_A NFC_VDD_D NFC_VDD_RF NFC_VDD_AM NFC_VDD_DR NFC_AGDC GND
EXPECTED RESULT	each regulator rail carries its specified parallel pair
VISUAL DELTA	five new decoupling legs in the cap row
HOW TO CHECK	each rail carries exactly the IC pin and its capacitors and nothing else - no rail back-driven, which is D-047's load-bearing half and it currently holds

45

RQ-029

REPAIR

DATASHEET-DERIVED

P2

nfc-host

one part leaves the BOM

OBJECTS

R43-NFC (4.7 k)

NETS

NFC_IRQ 3V3

EXPECTED RESULT

one part leaves the BOM

VISUAL DELTA

one body disappears

HOW TO CHECK

NFC_IRQ still reaches U12-NFC.27 and the ESP32-S3 receiver after removal

46

RQ-030

REPAIR

BOM/CPL

P1

audio-pdm

As drawn, anyone populating R40/R41 to run the experiment contracts/audio-interface.md says 'must be proven' places 10 kOhm in series with a 3.072 MHz PDM clock and its return data. The RC time constant destroys the edge, and the failure presents as a firmware or decimation problem rather than a BOM problem. R38/R39 (route A, fitted) are correctly RC0402FR-070RL, 0 Ohm.

OBJECTS

R40-AUD R41-AUD

NETS

PDM_CLK PDM_CLK_RT PDM_DAT PDM_DAT_RT

EXPECTED RESULT

the documented route-B experiment becomes performable; the DNP alternates stop naming a 10 kOhm part

VISUAL DELTA

two value texts change

HOW TO CHECK

part becomes 0 Ohm or 22 Ohm (matching DEC-06), keeping the DNP fitting instruction

47

RQ-031

REPAIR

DATASHEET-DERIVED

P1

audio-pdm

blocked · DEC-06

after RQ-030

The XOR cannot be enforced by topology and it is honest to say so: the microphone has one clock pin, so any two candidate drivers must meet there. PDM_CLK/PDM_DAT are the minimum achievable merge point and no 0R/DNP matrix can do better. The per-owner nets ARE distinct (PDM_CLK_ADC vs PDM_CLK_RT, PDM_DAT_ADC vs PDM_DAT_RT). The asymmetry matters: both clock resistors populated hard-shorts the ADC6120 GPIO1 OUTPUT to the RT1062 SAI clock OUTPUT through 0 Ohm; the data case is two receivers and is benign. Existing note e129958 states which route is default but does not state 'never both'.

OBJECTS

R38-AUD R39-AUD R40-AUD R41-AUD sheet text e129958

NETS

PDM_CLK PDM_DAT

EXPECTED RESULT

double-population becomes current-limited instead of a dead short; the sheet note prohibits the both-populated case

VISUAL DELTA

up to four values change; one text block is rewritten

48

RQ-032

REPAIR

DATASHEET-DERIVED

P1

audio-mic

blocked · DEC-07

Three problems in one net: the name inverts the behaviour (writing 1 to 'MIC power enable' turns the microphone off); the default is ON through RT1062 reset, ISP/recovery and permanently if firmware never boots; and all 22 TEXT primitives on the sheet were read and none mentions MIC_PWR_EN, its polarity, or the rail's power-on state, which both audio contracts require.

OBJECTS

R9-PWR2 (100 k) net MIC_PWR_EN a new sheet annotation

NETS

MIC_PWR_EN 3V3_MIC_REG GND

EXPECTED RESULT

the mic rail's power-on state becomes a chosen state rather than one defined by accident

VISUAL DELTA

one resistor's far end moves; one net label may change; one text block appears

HOW TO CHECK

if DEC-07 selects default-OFF, R9 pulls UP to 3V3_MIC_REG - Q1's SOURCE, never the main 3V3 rail, so Vgs can never reverse during rail sequencing. Sheet annotation reads 'MIC_PWR_EN: ACTIVE LOW (P-FET gate) | default <ON|OFF> via R9 pull-<down|up>'. If the default stays ON, rename the net MIC_PWR_EN_N.

49

RQ-033

WIRE

DATASHEET-DERIVED

P2

audio-flex

[INFERENCE] pins 11/12 are the hold-down / shell tabs: a 10-position part presented with 12 symbol pins, and the sheet's own annotation e129957 accounts for signal pins 1-10 only. An ungrounded shell on a flex carrying a 3.072 MHz clock is an EMI liability in both directions and gives the flex shield no return. The GND-interleaved signal ordering on pins 1-6 is otherwise correct.

OBJECTS

J9-AUD.11 J9-AUD.12

NETS

GND

EXPECTED RESULT

the flex connector shell gains a return

VISUAL DELTA

two short wires at the FH12-10S-0.5SH

HOW TO CHECK

source read-back: pins 11 and 12 on GND

50

RQ-034

REPAIR

DATASHEET-DERIVED

P2

audio-adc

blocked · DEC-12

On a board whose stated purpose is resolving 113 dB against 123 dB, three floating high-impedance analogue nodes on the measurement die are an avoidable noise-injection path. All other ADC6120 decoupling verifies correct against the datasheet: DREG 10 uF + 0.1 uF, AREG 10 uF + 0.1 uF, VREF 1 uF, EP grounded.

OBJECTS	U11-AUD.1 (IN1P) U11-AUD.2 (IN1M) U11-AUD.4 (IN2M_GP01)
NETS	AVSS or a common-mode reference
EXPECTED RESULT	three high-impedance analogue nodes stop floating on the measurement die
VISUAL DELTA	three wires, or three NC flags plus a sheet annotation
HOW TO CHECK	either terminated per the datasheet, or registered explicitly as GPIx/GPOx via CHx_INSRC[1:0] with that decision recorded on the sheet

51

RQ-035

REPAIR

DATASHEET-DERIVED

P1

audio-clock

blocked · DEC-13

The procedure as written is electrically safe and self-defeating. Removing R31-R33 does isolate the RT's outputs and the named resistors are the correct ones (the cut nearest the RT, leaving no long unterminated stub). But it also leaves the RT with no BCLK and no FSYNC, so it can no longer sample AUDIO_DOUT (U11-AUD.6 -> R37 22 Ohm -> U6-RTC.H12). contracts/audio-interface.md requires the capture path used for dynamic-range evaluation to preserve full sample width, which puts the SAI INSIDE the measurement loop. No 16-bit path is imposed by the hardware - SDOUT is a plain TDM/I2S output and nothing on the sheet truncates it.

OBJECTS	sheet text e129956 R31-AUD R32-AUD R33-AUD
NETS	AUDIO_MCLK_RT AUDIO_BCLK_RT AUDIO_FSYNC_RT
EXPECTED RESULT	the documented external-clock procedure stops silently disabling capture through the RT1062
VISUAL DELTA	one text block is rewritten
HOW TO CHECK	per DEC-13, the note either reads 'leave R31-R33 fitted; configure the RT1062 SAI as clock slave; drive J8' with the 22 Ohm acting as damping, or it keeps the removal procedure AND states that capture through the RT is unavailable in that mode

52

RQ-036

REPAIR

DATASHEET-DERIVED

P1

esp-service

As drawn the divider gives 2.500 V at 5.00 V nominal - 25 mV of margin, INSIDE the tolerance of the resistors alone - and 2.375 V at 4.75 V (USB 2.0 minimum at a downstream port), which READS LOW: the S3 believes the service port is unplugged while it is plugged in. The whole self-powered-USB detection story annotated on the sheet rests on this node. The sense-only topology itself is PROVEN correct: S3_VBUS reaches exactly six pins with no diode, FET, load switch, ferrite or resistor to any system rail.

OBJECTS	R72-ESP
NETS	ESP_USB_VBUS_SENSE GND
EXPECTED RESULT	the VBUS sense node clears VIH across the whole USB voltage range

VISUAL DELTA one value text changes from 100k to 150k

HOW TO CHECK with R71 at 100 k and R72 at 150 k: 3.00 V at 5.0 V VBUS, 2.61 V at 4.35 V (135 mV above VIH), 3.30 V at 5.5 V - still inside the VDD+0.3 V = 3.6 V absolute maximum. A small Schottky clamp to 3V3 is the alternative.

53

RQ-037-P PLACE DATASHEET-DERIVED P2 esp-power

component count +1

OBJECTS one series inductor or ferrite rated $\geq$ 500 mA

NETS

EXPECTED RESULT component count +1

VISUAL DELTA one new body in the module's 3V3 supply path

HOW TO CHECK placed in series between the board 3V3 net and U9-ESP pin 2

54

RQ-037-D DESIGNATE DATASHEET-DERIVED P2 esp-power after RQ-037-P

one new designator with rated current $\geq$ 500 mA recorded

OBJECTS the new LC element

NETS

EXPECTED RESULT one new designator with rated current $\geq$ 500 mA recorded

VISUAL DELTA designator and value text

HOW TO CHECK rated current 500 mA or above per Espressif

55

RQ-037-W WIRE DATASHEET-DERIVED P2 esp-power after RQ-037-D

The mic rail has FB5-AUD; the radio - the part that draws transmit bursts and radiates - has none. The S3's 355 mA Wi-Fi TX burst is 55% of the whole 3V3 peak and is irreducible: under D-007 the sole wireless control plane is BLE-MIDI, whose TX peak is 344 mA, so parking Wi-Fi does not materially reduce it.

OBJECTS U9-ESP.2 (3V3) the new LC element

NETS 3V3 a new filtered module rail

EXPECTED RESULT the radio's supply gains the vendor-specified LC

VISUAL DELTA the module's supply visibly passes through the new element

HOW TO CHECK source read-back: U9-ESP.2 no longer sits directly on the board 3V3 net

56

RQ-038 REPAIR DATASHEET-DERIVED P0 motion-i2c blocked · DEC-01 after RQ-048

THE LOAD-BEARING CORRECTION, and not the one the brief anticipated. Measured: on SDA and SCL, R44/R45 and R46/R47 bridge the IDENTICAL net pair - fitting either makes no electrical difference, so the pattern selects nothing. A link between MOTION_SDA and the shared bus isolates THE SENSOR, not THE MASTER. Bus census: four slaves (INA226, ADC6120, ST25R3916B, LIS2DH12), one master socket (ESP32-S3). The safety property the contract exists to enforce is satisfied VACUOUSLY - by the absence of the configuration the contract requires - which is worse than a violation because it looks clean. THE HAZARD ARMS ON REPAIR: once the split is real, a populated R45 ties the sensor to both masters, which is exactly the uncontrolled dual-master connection contracts/motion-interface.md forbids. So RQ-048 must land BEFORE or WITH this, never after. Pull-up arithmetic: R50-MOT 4.7 k currently pulls up MOTION_SDA; merged it lands parallel with R4-PWR1 and R28-AUD -> ~1.57 kOhm on SDA against 2.35 kOhm on SCL. MOTION_INT keeps its _RT/_S3 split through R48/R49 and is the one signal whose XOR is correctly placed already - an interrupt genuinely is point-to-point. MOTION_INT is NOT a dual-master risk: one push-pull output into two receivers is one driver into two loads.

OBJECTS	R44-MOT R45-MOT R46-MOT R47-MOT R50-MOT R4-PWR1 R28-AUD R29-AUD
NETS	MOTION_SDA MOTION_SCL I2C_SDA I2C_SCL RT_I2C_SDA RT_I2C_SCL S3_I2C_SDA S3_I2C_SCL
EXPECTED RESULT	a real owner-selection XOR exists for the first time; MOTION_SDA/MOTION_SCL collapse into I2C_SDA/I2C_SCL; two 0 Ohm pairs appear on the master legs; the pull-up budget is consolidated
VISUAL DELTA	R44-R47 move or are deleted and replaced on the master legs; two nets collapse; surplus pull-ups disappear
HOW TO CHECK	source read-back: each master leg carries one FIT and one DNP link to per-owner nets; the sensor sits directly on the shared bus; ONE pull-up pair remains on the shared bus, symmetric between SDA and SCL

57

RQ-039 REPAIR DRAWING-ONLY P2 motion-i2c after P-1

MOTION_INT_S3 binds exactly one pin; the nearest U9-ESP pin to e8944 is 90 units away. The DRC 'single network' warning is NOT a reliable dangling-net detector - MOTION_INT_S3 was a one-pin net in the frozen source and appeared in neither run's list. Its silence is not evidence.

OBJECTS	wire e8944, segment [4170,4365,4040,4365] R49-MOT.2
NETS	MOTION_INT_S3
EXPECTED RESULT	the S3 interrupt leg either terminates on a real pin or is deleted with the reason recorded
VISUAL DELTA	one wire lands on a pin, or one wire disappears

HOW TO CHECK

MOTION_INT_S3 reaches two pins, or it does not exist. CHECK FIRST whether the NC sweep flagged the intended receiving pin - if it did, the leg is now documented as permanently unused, which is worse than unterminated because it reads as intentional.

58

RQ-040-P

PLACE

DATASHEET-DERIVED

P3

motion

component count +1

OBJECTS

one 10 uF capacitor

NETS

EXPECTED RESULT

component count +1

VISUAL DELTA

one new body beside U13-MOT

HOW TO CHECK

placed local to U13-MOT pin 9

59

RQ-040-D

DESIGNATE

DATASHEET-DERIVED

P3

motion

after RQ-040-P

one new designator, value 10 uF

OBJECTS

the new bulk capacitor

NETS

EXPECTED RESULT

one new designator, value 10 uF

VISUAL DELTA

designator and value text

HOW TO CHECK

value 10 uF

60

RQ-040-W

WIRE

DATASHEET-DERIVED

P3

motion

after RQ-040-D

Hand to A2-RAILS if the 3V3 lane can show existing bulk within the datasheet's 'as near as possible' intent.

OBJECTS

U13-MOT.9 (VDD) the new capacitor

NETS

3V3 GND

EXPECTED RESULT

the accelerometer gets the second capacitor its datasheet asks for

VISUAL DELTA

one decoupling leg appears at VDD

HOW TO CHECK

source read-back: 3V3 at U13-MOT.9 carries both C62-MOT 100 nF and the new 10 uF

61

RQ-041-P

PLACE

DATASHEET-DERIVED

P1

led-output

component count +2

OBJECTS

two 10 k0hm 0402 pull-down resistors

NETS

EXPECTED RESULT component count +2

VISUAL DELTA two new bodies on the LED data path

HOW TO CHECK placed on LED_D0_3V3 and LED_D1_3V3, on the buffer-input side

62 **RQ-041-D** DESIGNATE DATASHEET-DERIVED P1 led-output after RQ-041-P

two new designators, value 10k

OBJECTS the two new pull-downs

NETS

EXPECTED RESULT two new designators, value 10k

VISUAL DELTA designator and value text

HOW TO CHECK value 10 kOhm; bound to a device whose value census is 10 k (not the shared mixed-value device)

63 **RQ-041-W** WIRE DATASHEET-DERIVED P1 led-output after RQ-041-D

Measured: LED_OE_L binds exactly two pins (U14-LED.1, R53-LED.1) with NO dangling stub - no controller reaches either OE pin - and R53/R54 are 10 k pull-DOWNS to GND, so OE is hard-tied ENABLED. There is no pull-down anywhere on LED_D0_3V3, LED_D1_3V3, LED_D0_J or LED_D1_J. Between power-on and RT firmware taking control the strip data state is indeterminate: random pixel garbage at best, a sustained high-current white frame at worst. This is LED-audit Option A, recommended - it costs no RT pads and no firmware sequencing.

OBJECTS LED_D0_3V3 LED_D1_3V3 the two pull-downs

NETS LED_D0_3V3 LED_D1_3V3 GND

EXPECTED RESULT both buffer inputs are held low until the RT drives them; the strip data line is defined through power-on and reset

VISUAL DELTA two pull-down legs appear on the data path

HOW TO CHECK source read-back: each LED data net reaches the RT1062 pad, the buffer input and the new pull-down

64 **RQ-042-P** PLACE DATASHEET-DERIVED P1 led-thermal

component count +2

OBJECTS two NTC bias resistors

NETS

EXPECTED RESULT component count +2

VISUAL DELTA two new bodies at the NTC nodes

65

RQ-042-D

DESIGNATE

DATASHEET-DERIVED

P1

led-thermal

blocked · DEC-09

after RQ-042-P

two new designators with the value chosen in DEC-09

OBJECTS

the two bias resistors

NETS

EXPECTED RESULT

two new designators with the value chosen in DEC-09

VISUAL DELTA

designator and value text

HOW TO CHECK

value chosen so the NTC sits mid-span across the intended monitoring range; 10 k is the conventional match to a 10 k-at-25 C NTC but the choice belongs to the thermal lane (D15)

66

RQ-042-W

WIRE

DATASHEET-DERIVED

P1

led-thermal

after RQ-042-D

Measured: LED_THERM_L binds exactly two pins (RT1-LED.1, U6-RTC.L12) and LED_THERM_R two (RT2-LED.1, U6-RTC.K12); the NTCs' other terminals are on GND. With no top leg there is NO EXCITATION - the ADC reads an undriven node, not a temperature - and the failure is silent: firmware gets a plausible-looking number that means nothing. This audit asserts only that the resistor must exist.

OBJECTS

RT1-LED RT2-LED the two bias resistors U6-RTC.L12 U6-RTC.K12

NETS

LED_THERM_L LED_THERM_R 3V3

EXPECTED RESULT

both NTC dividers come into existence

VISUAL DELTA

two bias legs appear from 3V3 to the thermal nodes

HOW TO CHECK

source read-back: each LED_THERM net reaches the NTC, the bias resistor and the RT1062 ADC ball

67

RQ-043

REPAIR

DATASHEET-DERIVED

P2

led-output

blocked · DEC-10

after RQ-041-W

Execute only if DEC-10 selects option B. B does NOT remove the need for RQ-041's pull-downs - it adds to them, because a Hi-Z buffer still leaves the strip data line undefined. Costs two RT pads.

OBJECTS

R53-LED R54-LED U14-LED.1 U15-LED.1 two RT1062 GPIOs

NETS

LED_OE_L LED_OE_R +5V_LED_L +5V_LED_R

EXPECTED RESULT

the buffers boot Hi-Z and firmware enables them explicitly

VISUAL DELTA

two resistors change rail; two new wires leave the LED block toward the RT1062

68

RQ-044

REPAIR

DATASHEET-DERIVED

P1

led-power

blocked · DEC-14

A ferrite is an EMI element, not a protection device, and not a switch. A short on one strip pulls the shared eFuse down and takes BOTH channels out. The three No-Connects on U4-PWR2 (PG, PGTH/FLT, ITIMER) are LEGITIMATE - TI characterises the device with all four open - but the consequence is a product-level risk: the fitted part is the L suffix, LATCHED OFF, EN/UVLO is hard-tied to 5V_SYS, FLT is unconnected, and with ITIMER open the blanking is at minimum. An inrush transient from hot-plugging a strip latches the branch off, cannot be cleared by firmware, requires a full input power cycle, and gives firmware no signal explaining why the strips went dark. Soft-start is fine: C68-PWR2 2.2 nF gives ~0.91 V/ms, ~5.5 ms ramp. A2-RAILS owns the device choice.

OBJECTS

FB1-PWR2 FB2-PWR2 U4-PWR2.4 (PGTH/FLT)

NETS

5V_LED_COMMON +5V_LED_L +5V_LED_R

EXPECTED RESULT

each LED branch becomes independently protectable, sheddable and observable

VISUAL DELTA

two ferrites are replaced by or joined with switch/protection devices; one new wire to the RT1062

HOW TO CHECK

each branch carries a real protection/switch element; FLT reaches an RT1062 input

69

RQ-045

REPAIR

DATASHEET-DERIVED

P1

led-power

blocked · DEC-02

The LED branch is 46% of the sustained trunk and 51% of the peak, and it is unconditionally enabled at power-on before any firmware runs. On a Default-USB source the board attempts roughly 4x the permitted current and the source folds back or shuts down.

OBJECTS

U4-PWR2.1 (EN/UVLO) an RT1062 GPIO

NETS

5V_SYS a new LED enable net

EXPECTED RESULT

the board's largest load becomes sheddable in firmware

VISUAL DELTA

one wire moves from the hard tie to a processor pin

HOW TO CHECK

source read-back: U4-PWR2.1 no longer sits directly on 5V_SYS

70

RQ-046

REPAIR

BOM/CPL

P0

bom

Proved twice by two independent implementations compared in emitted order, disagreeing nowhere. The decisive row is the COMPONENT record's own library reference: ["COMPONENT";"e426";"RC0402FR-0710KL.1",365,4420,...] for R1, drawn as 1.33k. At 10 kOhm the trunk eFuse limits to 0.300 A guaranteed against a 2.08 A peak and a 1.15 A sustained load:

THE BOARD CANNOT POWER UP - the 3V3 branch alone draws 0.480 A at peak, 1.6x the limit, so the eFuse trips during the buck's own start-up. It presents at bring-up as 'the board is dead' with no obvious cause. ERC and DRC cannot see it: both check connectivity, and this is a data defect in a field neither inspects. See disagreement-4 for the open question of whether it would actually build wrong.

OBJECTS

R1-PWR1

NETS

EXPECTED RESULT R1 rebinds from device e1b1f220e40a4edea589adfa05a5d8c7 (24 instances, dominant identity RC0402FR-0710KL = 10 kOhm) to device 31e09015057d40e885a10cd7f4784b79 (C276261, RC0402FR-071K33L); supplierId set to the LCSC code

VISUAL DELTA none on the canvas; the value text changes only if RQ-021 runs in the same pass

HOW TO CHECK device binding, MPN, supplier code and displayed value ALL agree on the value chosen in RQ-021

71

RQ-047

REPAIR

BOM/CPL

P0

bom

Changing a value that verifies correct would be churn. Only the binding is wrong.

OBJECTS

R8-PWR2

NETS

EXPECTED RESULT R8 rebinds to device 62850562b7904f11a759e60c3de20f54 (C185418, RC0402FR-073K48L); supplierId set to the LCSC code

VISUAL DELTA none

HOW TO CHECK all four BOM fields agree on 3.48 kOhm. DO NOT CHANGE THE VALUE - 3.48 kOhm is correct and verified against the envelope: 0.958 A typ / 1.054 A max, safely under the repaired trunk's 2.42 A guaranteed trip, and inside TI's recommended R_ILM range of 549 Ohm to 6.65 kOhm.

72

RQ-048

REPAIR

BOM/CPL

P0

bom

RC0402FR-07DNP DOES NOT EXIST - no LCSC result, and not a valid YAGEO RC0402 part number. The bind map confirms it is a deliberate label pointing at the real 10 kOhm device, and all seven COMPONENT records carry library ref RC0402FR-0710KL.1. As the BOM stands they will be manufactured and fitted as 10 kOhm resistors at locations the schematic says must be empty. The mechanism is not missing - it is applied correctly to C43-ESP, C44-ESP and C52-AUD, and was mutation-tested (stripping it from those three makes all three flag; a FITTED part never flags). MUST LAND BEFORE OR WITH RQ-038: a populated R45 arms the dual-master condition the moment the ownership matrix becomes real. Also: a 10 kOhm in

series on an I2C line does not select an owner, it breaks the bus - against 4.7 kOhm pull-ups it forms a divider and neither side meets V_OL.

OBJECTS	R40-AUD R41-AUD R45-M0T R47-M0T R49-M0T R56-VAL R57-VAL
NETS	
EXPECTED RESULT	seven parts gain Add into BOM = no AND Convert to PCB = no; the fabricated MPN RC0402FR-07DNP is cleared
VISUAL DELTA	none
HOW TO CHECK	BOTH flags, not one - BOM exclusion alone leaves a CPL row with no BOM line, which JLC flags (verify_fab_package.py check C2). Confirm against RQ-056's real export.

73

RQ-049 REPAIR NETLIST-CONFIRMED P1 bom

TWO connectors, not one. The symbol exposes a single shell pin (EH, pin 1); the footprint carries four shell/hold-down pads, so pads 2-4 will be left netless in layout. This is a schematic-side symbol/footprint mismatch, so it belongs to VAL-G2 rather than VAL-G3.

OBJECTS	J1-PWR1 (\$1I339) J7-ESP (\$1I8334) the USB4105-GF-A symbol
NETS	
EXPECTED RESULT	the shield/retention tabs on both receptacles become nettable
VISUAL DELTA	the symbols gain pins
HOW TO CHECK	DRC line 430 clears. Rebind to a symbol exposing the shell pins, or add shell pins to the current symbol, then net them (typically chassis/GND through the intended shield strategy). DO NOT resolve by deleting the pads.

74

RQ-050 REPAIR NETLIST-CONFIRMED P2 bom after RQ-056

A defect only the DRC found - missed by every lane. EasyEDA independently confirms U6-RTC is ONE multi-part component, settling the two-MCU question from the tool itself. The Add into BOM = no on part 2 is almost certainly intentional de-duplication; if EasyEDA's BOM engine already de-duplicates, the override is redundant and the warning is its only effect.

OBJECTS	U6-RTC parts \$1I3295 and \$1I3673
NETS	
EXPECTED RESULT	Supplier Part, Add into BOM and supplierId agree across both parts of the multi-part symbol; DRC line 417 clears
VISUAL DELTA	none
HOW TO CHECK	export the BOM and confirm the MCU appears EXACTLY ONCE; if it does with the Add into BOM = no override removed, remove it

75

RQ-051 DESIGNATE BOM/CPL P1 bom blocked · DEC-15

Not errors of execution - someone chose each one - but BLOCKER anyway by the same test: the part that gets fitted is not the part the schematic displays, and nothing on the sheet says so. A note in a side file is not a declaration; a fabricator never sees it. The two headers are a DIFFERENT FOOTPRINT FAMILY - plain 0.1 inch header against shrouded JST-XH - and must be checked before layout, not after. R42-NFC additionally needs DEC-15: a 5% resistor in an NFC matching network is an electrical choice, not a sourcing convenience.

OBJECTS R42-NFC J6-ESP J11-VAL

NETS

EXPECTED RESULT three undeclared stand-ins gain an explicit waiver visible on the sheet

VISUAL DELTA three text annotations appear beside the parts

HOW TO CHECK each waiver names the substituted part. R42-NFC: drawn RC0402FR-072R2L (2.2 Ohm 1% FR), bound RC0402JR-072R2L (5% JR). J6-ESP and J11-VAL: drawn PREC006SAAN-RC, bound HEADER_MALE-XH_1X6P_2.54MM.

76

RQ-052 REPAIR BOM/CPL P2 bom blocked · DEC-16

All eight carry Manufacturer Part = 5001 (Keystone 5001, a real purchased through-hole part) while bound to EasyEDA's generic TP-yellow-testpoint device, a bare pad.

OBJECTS TP1-ESP TP2-ESP TP3-AUD TP4-AUD TP5-AUD TP6-VAL TP7-AUD TP8-AUD

NETS

EXPECTED RESULT eight phantom BOM lines resolve one way or the other

VISUAL DELTA none

HOW TO CHECK MPN and binding agree. If bare pads: clear the 5001 MPN and set Add into BOM = no. If real Keystone parts: bind the real device.

77

RQ-053 REPAIR BOM/CPL P1 bom after RQ-046, RQ-047, RQ-056

160 MAJOR instances are ONE finding repeated, not 160 mistakes - the correct part is bound, its code simply has to be resolved. Precedent: IM69D130 raw-to-normalised export differs in exactly one respect, Supplier Part (0402WGF2201TCE.1 -> C25879), nothing else changing. EasyEDA's own DRC already reports it at scale: 209 of 228 designators (92%) flagged 'Component attributes does not match the Supplier Part'. The 19 clean ones are C10-PWR2, C35/C36-RTDBG, C42/C43/C44-ESP, C51/C52-AUD, C68-PWR2, C92-C97-NFC, L1-PWR2, L4-RTC, R75-PWR2, R76-NFC. Also empty on all 229: LCSC Part Name, Supplier Footprint, JLCPCB Part Class.

OBJECTS	172 instances whose supplierId holds an EasyEDA device key of the form <MPN>.<n>
NETS	
EXPECTED RESULT	supplierId carries a real LCSC code on every instance; JLC can auto-match the board
VISUAL DELTA	none
HOW TO CHECK	complete the device-to-LCSC table FIRST - only 8 of 61 bind-map entries carry an lcsc field, and that is the real work in this item. Then run the normalisation already proven on the IM69D130 fab package.

78

RQ-054 REPAIR BOM/CPL P2 bom

The FOOTPRINT is justified - it damps the fast AHCT edge into an unterminated strip lead, which is exactly what it is for. The VALUE is not: series termination is a function of driver output impedance and routed trace and lead geometry, none of which exists yet. Set at VAL-G3 from the routed geometry. Do not freeze it at this gate.

OBJECTS	R51-LED R52-LED
NETS	
EXPECTED RESULT	the series data resistors stop being frozen at a donor-circuit number
VISUAL DELTA	two value texts change from 33R to TUNE_TBD
HOW TO CHECK	value field reads TUNE_TBD in both BOM and schematic; recorded as VAL-G3-LED-01

79

RQ-055 REPAIR BOM/CPL P3 bom

100 pF of shunt on a USB data line is an order of magnitude above what a tuning cap should be; 100 pF on an MCLK line up to 24.576 MHz is a large load. All three are correctly DNP today and correctly carry Add into BOM = no - the note is that the alternate VALUE must be re-picked before anyone fits them. The tuning footprints themselves are good practice and match Espressif's reference topology.

OBJECTS	C43-ESP C44-ESP C52-AUD
NETS	
EXPECTED RESULT	the stamped DNP alternates stop naming a value that would break the line if fitted
VISUAL DELTA	three value texts change from 100pF to a few pF or blank
HOW TO CHECK	alternate value re-picked (~10 pF is the usual starting point); DNP fitting instruction retained

80

RQ-057 REPAIR DRAWING-ONLY P1 esp-service

ONE systematic placement offset, not seventeen errors - one batch stub-translation fixes it. Root cause dated and attributed: the rejected-but-never-rolled-back canonical-esp-service-label-visibility-repair at 474325:91295516, whose first repair aborted with no change and whose second was ACCEPTED while changing only metadata. Canon K1E-041. DO NOT RE-RAISE AS AN ELECTRICAL DEFECT - see withdrawn-3.

OBJECTS	~17 U9-ESP lead-in wire stubs
NETS	GND 3V3 ESP_EN NFC_IRQ MOTION_INT_S3 S3_P0R_REQ RT_PWR_VALID ESP_USB_VBUS_SENSE K1BR_SCK K1BR_M0SI K1BR_MISO_S3 K1BR_CS K1BR_IRQ_S3 USB_DP_S3 USB_DM_S3 ESP_UART0_RX ESP_UART0_TX I2C_SDA I2C_SCL
EXPECTED RESULT	drawing geometry lands; NO netlist change expected
VISUAL DELTA	~17 stubs move by one sub-grid step onto their pins
HOW TO CHECK	verify by READ-BACK, not by eye. The module's pin cloud sits at (+5, -20) from its own anchor; the left column shows a repeating (+5, 0) and the K1BR bus a (-5, +20). Afterwards the 14:58 floating count for U9-ESP must remain ZERO - a change there would mean this transaction broke something.

81

RQ-058 REPAIR DRAWING-ONLY P2 power-buck

C10's pin cloud sits at delta (0.0, 0.0) from its own source anchor - THE PIN DATA IS RIGHT AND THE WIRE IS 5 UNITS OFF. This is the measured counterexample that killed the DRC-fitted registration method: applying a (0,-5) 'correction' would have moved good pin data onto a badly drawn wire and erased the defect - on the buck soft-start capacitor, one of the parts D-045 covers.

OBJECTS	C10-PWR2's GND stub
NETS	GND
EXPECTED RESULT	drawing geometry lands
VISUAL DELTA	one 5-unit stub lengthens to the sheet's standard 20 units
HOW TO CHECK	the stub reaches C10-PWR2 pin 2 at page (1040, 4295)

82

RQ-059 REPAIR DRAWING-ONLY P2 esp-service after RQ-057

The 10 k pull-up R62-VAL is correct and inside SBVS050M's 10 kOhm-1 MOhm window. Pair with RQ-011: wiring SENSE without a consumer, or a consumer without SENSE, still leaves RT_PWR_VALID decorative. May be subsumed by RQ-057.

OBJECTS	wire e8950
NETS	RT_PWR_VALID
EXPECTED RESULT	drawing geometry lands on U9-ESP pin 9

VISUAL DELTA	one stub lengthens by +5 in x, half a pin pitch
HOW TO CHECK	RT_PWR_VALID reaches R62-VAL.2, U16-VAL.1 and U9-ESP.9

83

RQ-060 REPAIR DRAWING-ONLY P2 sheet-wide after RQ-003, RQ-057

Several of these belong to the same U9-ESP displacement - run RQ-057 first. Do not treat a resolved label count as an electrical claim: the checker states in its own header that it is NOT a netlist oracle.

OBJECTS	8 net labels meeting fewer than two pins 25 wires meeting no pin
NETS	BUCK_PG ESP_UART0_RX K1BR_IRQ_S3 K1BR_MISO_S3 K1BR_MOSI LED_D0_3V3 MOTION_INT_S3 S3_P0R_REQ
EXPECTED RESULT	check_schematic_connectivity.py moves toward GREEN
VISUAL DELTA	~33 stubs land on pins
HOW TO CHECK	STATE THE SNAP TOLERANCE AND SHOW THE SENSITIVITY, or it is not a verdict. Raw lands 654/881 pins (74%); +/-5 units reaches 669/881 (76%) and drops the sub-two-pin label count from 7 to 4. The checker refuses any tolerance >= half the measured pin pitch and discovers the pitch from the data.

84

RQ-061 WIRE DRAWING-ONLY P2 power-tree

Both answers about the power tree are true. ELECTRICALLY it is a complete, correct tree and it passes K1E-017: every branch has a real active source, and NFC_5V is a legitimate ferrite filter from 5 V to 5 V, not a rail conjured out of nothing. VISUALLY it is not: zero of the 675 wires touch any other wire anywhere on the sheet - 5V_SYS is 12 disjoint wires and 3V3 is 90. The contract permits stub-and-label for global rails, so this is a readability finding, not an electrical one - but POWER-ARCHITECTURE.md singles out this exact spine ('six disconnected 5V_SYS labels are not a power tree') and that specific request is unmet.

OBJECTS	the eight trunk hops inside PWR1/PWR2 only
NETS	5V_USB 5V_USB_FILTERED 5V_PROTECTED 5V_SYS 3V3 5V_LED_COMMON NFC_5V 3V3_MIC_REG
EXPECTED RESULT	the power spine becomes visible wiring rather than disjoint labels; roughly 12-15 wire additions in one region
VISUAL DELTA	the trunk becomes a drawn tree at whole-sheet scale - entry to eFuse to 5V_PROTECTED to shunt to 5V_SYS to the four branch heads
HOW TO CHECK	the eight hops are continuous wiring. DO NOT attempt to de-label the 90-wire 3V3 net; that is not what POWER-ARCHITECTURE.md asks for and it would make the sheet worse. Leave leaf decoupling on labels.

85

RQ-062

DESIGNATE

DRAWING-ONLY

P3

sheet-wide after P-1, RQ-004, RQ-005, RQ-006

The 13 S3 spare GPIO are genuinely unconnected - none is a strapping pin, none is reserved by the octal PSRAM, none has an assigned function. The three PSRAM-reserved pins (IO35/36/37 on module pins 28/29/30) and the three strapping pins (IO3, IO46, IO45) are correctly left unconnected and MUST STAY THAT WAY - do not flag or pull those. For the tact switches the better fix is to parallel pins 3/4 onto 1/2, the normal PTS645 treatment, which also halves contact resistance.

OBJECTS

U1-PWR1.10 (ITIMER) U6-RTC.N12 (CHD_B) U6-RTC.N7/.P6/.P7 (OTG2)
 J1-PWR1.A8/.B8 J7-ESP.A8/.B8 U16-VAL.3 (CT) U16-VAL.4 (MR)
 SW1/SW2/SW3/SW4 unused throws U9-ESP spare GPIO IO16, IO17,
 IO18, IO8, IO9, IO21, IO47, IO48, IO38, IO39, IO40, IO41, IO42

NETS

EXPECTED RESULT intent becomes recorded rather than inferred; ~30 pins move from FLOATING to INTENTIONAL_NC

VISUAL DELTA ~30 NC crosses appear

HOW TO CHECK RUN THIS AFTER RQ-004/005/006 AND READ P-1's RESULT FIRST - this transaction must not re-flag a pin the queue is about to wire. Only 12 NC marks exist on the sheet today.

86

RQ-063

DESIGNATE

DRAWING-ONLY

P2

nfc-rf blocked · DEC-04 after P-1

U12-NFC.23 is the ONLY floating pin on U12-NFC at 14:58 - exactly what AN5592's single-ended stage predicts. In the frozen source BOTH pin 15 and pin 23 touched zero wires and the NC transaction marked NEITHER (its applied.json lists pins 2, 17, 18, 19, 25, 28, 29, 31 only). At 14:58 pin 23 is still floating and pin 15 is not, so one half of the differential pair has been acted on while the topology ruling is still open.

OBJECTS

U12-NFC.23 (RFI2) U12-NFC.15 (RF02)

NETS

EXPECTED RESULT the single-ended topology becomes a recorded decision on the canvas

VISUAL DELTA one or two NC crosses

HOW TO CHECK DO NOT FLAG EITHER PIN BEFORE DEC-04 IS RULED. An NC mark freezes the topology choice before the evidence is ratified - which is precisely what NFC-D16 flagged. First have the single writer report the LIVE state of U12-NFC.15: wired, NC-marked, or unchanged, and under what authority it changed.

87

RQ-064

REPAIR

DRAWING-ONLY

P3

k1br

ELECTRICALLY EVERY LINE IS CORRECT - a 22 Ohm series resistor between the two MCUs, with IO10-IO13 the correct ESP32-S3 default FSPI pins (FSPICS0/FSPICLK/FSPID/FSPIQ) for slave operation, and the forbidden-payload list satisfied at the physical layer: no RAW_PCM, RAW_PDM, AUDIO_FEATURES, RENDER_BUFFER, PIXEL_BUFFER, CRGB or any AUDIO_*, PDM_* or LED_* net touches any U9-ESP pin. But three nets use _RT and two use _S3, so the bare name means 'S3 side' for CS/SCK/MOSI and 'RT side' for MISO/IRQ, and a netlist diff will mis-attribute sides. TP1/TP2 tap only SCK and CS, so the seam cannot be fully observed on a scope - a gap against contracts/debug-fabric.md's purpose for a validation board.

OBJECTS	K1BR_CS_RT K1BR_SCK_RT K1BR_MOSI_RT K1BR_MISO K1BR_IRQ K1BR_CS K1BR_SCK K1BR_MOSI K1BR_MISO_S3 K1BR_IRQ_S3 three new test points
NETS	all five K1BR signals
EXPECTED RESULT	net naming becomes unambiguous (K1BR_<sig>_RT / K1BR_<sig>_S3 on all five); MOSI, MISO and IRQ gain test access
VISUAL DELTA	ten net labels change; three test points appear
HOW TO CHECK	every K1BR net carries an explicit side suffix; each of the five lines has a test point

88

RQ-065 REPAIR DRAWING-ONLY P3 nfc-symbol

Electrically inert. Recorded because a symbol that disagrees with its datasheet is how the next reader gets it wrong.

OBJECTS	U12-NFC symbol pins TAD1 and TAD2
NETS	
EXPECTED RESULT	the symbol becomes datasheet-faithful
VISUAL DELTA	two pin names swap
HOW TO CHECK	names match ST DS13541 Rev 5 Table 2

89

RQ-066 DESIGNATE DRAWING-ONLY P3 sheet-wide blocked · DEC-04

contracts/nfc-interface.md carries no VDD/VDD_TX co-supply rule, no RFI attenuation requirement, no EMC cutoff window and no single-drive register bit (NFC-D9). Fixed 2.2 MHz with no spread spectrum beside a 13.56 MHz NFC front end is a deliberate EMC choice that should be recorded, not rediscovered at test.

OBJECTS	new sheet text blocks
NETS	
EXPECTED RESULT	design intent stops being rediscoverable only by re-derivation
VISUAL DELTA	several text blocks appear

records (a) every NFC matching value stamped provisional; (b) the single = 1 / rfo2 = 0 firmware requirement; (c) the 3V3-not-VDD_D I2C_EN strap variance with its rating justification; (d) the S-CONF to VIN EMC choice - 2.2 MHz, spread-spectrum OFF, discharge OFF, matched to the 2.2 uH L1-PWR2

Rulings only you can make

16 decisions that belong to Captain, not to a writer. 6 repairs cannot start at all until these land.

DEC-01

Does the RT1062 get its own I2C path to the ADC6120, or does the S3 configure it over K1BR?

RECOMMENDATION ESCALATED WITHOUT A RECOMMENDATION - the audio lane raised it as 'a decision, not a chore' and refused to resolve it.

EVIDENCE Measured: U6-RTC is on NEITHER I2C_SDA NOR I2C_SCL. The RT1062 has no I2C access anywhere on this board. The bus is four slaves (INA226, ADC6120, ST25R3916B, LIS2DH12) and ONE master socket - the ESP32-S3. The RT1062 does own the ADC6120's data and clock pins; it does NOT own and cannot reach its control port - sample rate, channel configuration, PDM mode, PLL setup, gain and the GPIO1/GPI1/GPI2 function assignments all live behind I2C. So every audio experiment on this validation board requires the ESP32-S3 to be alive and to configure the ADC on the RT1062's behalf, with the request crossing K1BR - on a board whose stated purpose includes resolving a 113 dB versus 123 dB converter question. Either a deliberate split (S3 = configuration master, RT = data master) that the ownership matrix must be updated to state, or an oversight. Separately, contracts/motion-interface.md names the RT1062 as default owner and the RT1062 has no data path to the sensor at all. Isolation IS achievable - select the master leg, not the sensor leg - but there is no RT master leg to select until two RT1062 balls are assigned.

BLOCKS RQ-038

DEC-02

The Type-C source requirement: 2.40 A at 5 V needs a source advertising 3.0 A, nothing measures CC, and the largest load cannot be shed.

RECOMMENDATION (a) SENSE AND THROTTLE - bring J1 CC1/CC2 to 5.1 kOhm Rd AND tap both to RT1062 ADC inputs through protection dividers, and put U4-PWR2.1 on an RT1062 GPIO. Costs two

ADC channels, one GPIO and four passives; fits the board's stated validation mission; and it is the only option that keeps the 2.40 A design point honest rather than aspirational.

EVIDENCE The board needs 12.0 W. USB Type-C R2.0 Table 4-18 gives three advertisements at 5 V: Default USB (500 mA, 2.5 W - 20% of requirement), 1.5 A (7.5 W - 62%), 3.0 A (15 W - the only one that covers it, 80% utilised). Three consequences, none addressed: (1) the board cannot read the advertisement - J1 has NO Rd at all so it never attaches, and J7's correct 5.1 kOhm Rds go straight to GND with no ADC tap, so nothing distinguishes a 15 W charger from a 2.5 W laptop port; (2) the board cannot shed its largest load - U4-PWR2.1 is hard-tied to 5V_SYS, so 51% of the peak is unconditionally enabled before firmware runs; (3) advertisement detection wants to live on J1, which is the connector with no CC circuit at all. Option (b) 'design to 1.5 A' does NOT survive the arithmetic: after the 1.030 A non-LED coincident peak the LED ceiling falls to roughly 0.15 A, which is not a lighting product - unless someone proves the Wi-Fi/BLE TX burst and the NFC TX burst cannot overlap, which is a firmware-scheduling claim, not a hardware one. And the trunk window carries only 2.4% of total slack ($3.000/2.397 = 1.252$ against the part's own 1.222 tolerance span), so exactly one standard ILIM value fits and any growth in the LED ceiling leaves NO SOLUTION AT ALL from a 3 A source. Option (c) 'specify the supply' is zero added circuitry and the weakest position for a validation board that will be plugged into whatever is on the bench.

BLOCKS RQ-045 RQ-017-P RQ-017-W

DEC-03 Which rail carries the NFC front end - both pins on NFC_5V, or both on 3V3?

RECOMMENDATION Not recommended by the lane; BOTH ARE LEGAL. 5 V gives full transmit drive with VDD_IO staying at 3V3 for host levels; 3V3 gives lower drive, a simpler PDN, and requires sup3V set in IO configuration register 2.

EVIDENCE Changes architecture/POWER-ARCHITECTURE.md and unsettles the 0.500 A NFC term in the trunk envelope. Whichever is chosen, NFC_5V keeps its ferrite and bulk.

BLOCKS RQ-026

DEC-04 Single-ended versus differential NFC antenna topology.

RECOMMENDATION RATIFY SINGLE-ENDED WITH CABLE (AN5592 S2.3) - which is what the board already is. Needs a decision-register entry and a contract line.

EVIDENCE

BLOCKS RQ-027-P RQ-027-D RQ-027-W RQ-063 RQ-066

DEC-05

Which rail does the TPS3808 supervisor monitor - 3V3 with the fitted G33, or 1V15_CORE with a part change to the adjustable G01 plus a divider?

RECOMMENDATION Monitoring 3V3 with the fixed-threshold G33 is the defensible reading of D13.1.

EVIDENCE The repair cannot dodge this. Even wired to 3V3 (as its sibling U7-RTC is), U16 reports on the shared 3V3 rail, not on anything RT-specific. The RT1062 core rail is 1V15_CORE, produced by the RT1062's own internal DCDC through L4-RTC. Monitoring the core needs a PART CHANGE, not a wire edit. RT_PWR_VALID is decorative at BOTH ends today - SENSE floats and no consumer is attached.

BLOCKS RQ-011

DEC-06

PDM route-A series value: keep 0 Ohm, or move to 22 Ohm to current-limit a double-populate?

RECOMMENDATION 22 Ohm - harmless at 3.072 MHz into a MEMS mic clock input, and already the house value elsewhere on this board.

EVIDENCE A real interlock would need a mux or a physically shared single footprint, which is a bigger change than this validation board warrants.

BLOCKS RQ-031

DEC-07

MIC_PWR_EN safe default: rail ON at power-up (as built), or OFF until firmware asks?

RECOMMENDATION Not recommended by the lane - but IF THE DEFAULT STAYS ON, rename the net MIC_PWR_EN_N so firmware cannot get the sense wrong.

EVIDENCE The name inverts the behaviour: writing 1 to 'MIC power enable' turns the microphone off. The current default is DEFINED, not indeterminate - but it was probably not CHOSEN. If the intended safe default is 'microphone unpowered until firmware asks for it', R9 must pull UP to 3V3_MIC_REG - Q1's source, not the main 3V3 rail, so Vgs can never be reversed during rail sequencing.

BLOCKS RQ-032

DEC-08

F1-PWR1: re-rate the trunk bead, or remove series filtering from the trunk entirely?

RECOMMENDATION Removal is worth serious consideration.

EVIDENCE Two independent problems wanting different fixes. Position: a ferrite in series with the whole rail, upstream of the eFuse and followed by C1-PWR1 22 uF, forms an LC tank across the hot-plug event, and the overshoot lands on the eFuse input whose OVLO is 5.46 V. Input EMI filtering generally does not belong in series with the full trunk ahead of the protection device.

BLOCKS RQ-019

DEC-09 NTC bias resistor value for LED_THERM_L / LED_THERM_R.

RECOMMENDATION The conventional match to a 10 kOhm-at-25 C NTC is 10 kOhm, but the value should put the NTC mid-span across the intended monitoring range - that choice belongs to the thermal lane (D15).

EVIDENCE The audit asserts only that the resistor must exist. Its absence is what makes both channels unreadable.

BLOCKS RQ-042-D

DEC-10

LED boot state: pull-downs on the data lines (option A), or OE pulled up plus firmware-commanded enable (option B)?

RECOMMENDATION A. Add B on top only if a firmware-commanded LED blackout is wanted for thermal or fault response - which, given the branch-protection gap, is a conversation worth having, but a separate one.

EVIDENCE B does not remove the need for A's resistors, it adds to them - a Hi-Z buffer still leaves the strip data line undefined. A costs no RT1062 pads and no firmware sequencing; B costs two RT pads.

BLOCKS RQ-043

DEC-11

U13-MOT.3 (SA0) strap - GND or 3V3 - and the resulting 7-bit address recorded in the D18 hardware-to-firmware contract.

RECOMMENDATION Strap deliberately either way; DO NOT leave it to the internal pull-up.

EVIDENCE The part will most likely answer at the SA0=1 address via its internal pull-up, but the strap is undeclared, noise-exposed, and the firmware contract has no address to bind to.

BLOCKS RQ-009

DEC-12 ADC6120 IN1P / IN1M / IN2M_GPO1: terminate, or register as GPIx/GPOx?

RECOMMENDATION Either, but RECORD THE DECISION ON THE SHEET.

EVIDENCE

BLOCKS RQ-034

DEC-13

External-clock override procedure: does the i.MX RT SAI support slave operation with BCLK/FSYNC as inputs and MCLK direction selectable?

RECOMMENDATION [INFERENCE - needs confirmation by whoever owns the RT1062 SAI configuration.] If yes, the correct procedure is 'leave R31-R33 fitted; configure the SAI as clock slave; drive J8', with the 22 Ohm acting as damping, and resistor removal becomes the fallback for the case where the SAI cannot be put into slave mode.

EVIDENCE The two contract requirements as currently realised are in direct conflict: the sheet's documented isolation procedure removes the RT1062's ability to receive, while contracts/audio-interface.md puts the RT1062 SAI inside the dynamic-range measurement loop.

BLOCKS RQ-035

DEC-14

LED per-branch protection and enable: which device replaces or joins FB1/FB2?

RECOMMENDATION A2-RAILS owns the device choice. The LED lane's finding is only that ferrite beads do not satisfy 'branch protection' or 'branch enable'.

EVIDENCE

BLOCKS RQ-044

DEC-15 R42-NFC: 5% JR stand-in, or the drawn 1% FR part?

RECOMMENDATION Not recommended.

EVIDENCE A 5% resistor in an NFC matching network is an electrical choice, not a sourcing convenience.

BLOCKS RQ-051

DEC-16 Test points: real Keystone 5001 parts, or bare pads?

RECOMMENDATION Not recommended; either way the MPN and the binding must agree, or eight phantom line items ship in the BOM.

EVIDENCE

BLOCKS RQ-052

Where two lanes disagree

Stated with both citations rather than silently resolved.

disagreement-1 Is U9-ESP electrically disconnected?

LANE A {'documents': ['USB-TOPOLOGY-AUDIT.md S3', 'PIN-AUDIT-PWR2.md PWR2-012'], 'position': '"0 of 41 pins connected / 36 of 41 have no wire; 'the highest-value repair on the sheet'"}

LANE B {'documents': ['anchors/schDrcLog_2026-08-28T1458.txt', 'PIN-AUDIT-NFC.md', 'RECONCILE-RECEIPT-2026-08-28b.md correction'], 'position': 'zero floating pins - fully connected'}

RESOLUTION Resolved by CLASS SEPARATION, not by averaging. The netlist is the authority for electrical claims on this host; the drawing defect is real against the single-sheet contract and is RQ-057. Both lanes are right about different properties. Lane A wrote before the 14:58 log existed.

disagreement-2

Was the U9-ESP registration offset real, or a measurement artefact to be fitted away?

LANE A {'documents': ['PIN-AUDIT-S3.md S1'], 'position': '"fitted a per-component translation (-5,-20) constrained to reproduce EasyEDA's DRC verdict; reports an exact fit on all 41 pins'"}

LANE B {'documents': ['ORACLE-BUILD-RECEIPT.md', 'RECONCILE-RECEIPT-2026-08-28b.md'], 'position': "DECLINED the method - two free parameters fitted to a target verdict will reproduce it; 221/228 is what the method PRODUCES, not evidence it is right, and it forfeits the DRC's value as an independent witness"}

RESOLUTION The orchestrator's instruction was WITHDRAWN from two other lanes on the strength of the counterexample: C10-PWR2 sits at delta (0,0) from its own anchor - the WIRE is 5 units off - so the 'correction' would erase the defect. A subagent's disagreement with the orchestrator is evidence, not friction. Recorded because the pattern matters more than the instance.

disagreement-3 How many RT1062 balls are unconnected - 111, 80, or 41?

LANE A {'documents': ['anchors/schDrcLog_2026-08-28.txt', 'PIN-AUDIT-AUDIO.md S3'], 'position': '111 at 12:17; geometry sees 38 with only 98 of 196 balls read back; 73 of the 111 lie outside that readback'}

LANE B {'documents': ['anchors/schDrcLog_2026-08-28T1458.txt'], 'position': '80'}

RESOLUTION UNRESOLVED AND UNADJUDICATED IN THIS QUEUE. PIN-AUDIT-RT.md did not land. A full 196-ball readback is required - every lane that touched the RT1062 asked for one. The 80 include L8 (USB D+), N6 (USB VBUS), N12, N7, P6, P7 and K6; the rest are believed unused GPIO where the DRC's 'place a No Connect Flag' advice is correct.

disagreement-4

Which field does EasyEDA actually order from, for R1-PWR1 and R8-PWR2?

LANE A {'documents': ['BOM-SEMANTIC-AUDIT.md S1.1'], 'position': '"both are ordered and fitted as 10 kOhm' - the COMPONENT library reference and the bound device both name RC0402FR-0710KL.1, and the correct LCSC code sits in Supplier Part, 'the field EasyEDA does not use for ordering'"}

LANE B {'documents': ['PIN-AUDIT-LED.md S5 correction', 'PIN-AUDIT-MOTION.md S4.4'], 'position': '"the field that carries the orderable identity is Supplier Part', so R8 orders as a genuine 3.48 kOhm and the LED limit is a determined 0.96 A'"}

RESOLUTION GENUINELY OPEN, and it changes the SEVERITY of RQ-046/RQ-047. Both readings agree the binding is wrong and must be repaired; they disagree on whether the board as it stands would build wrong. RQ-056 (export the BOM and read it) settles this and nothing else does. The repair is identical either way, so this does not block execution - only the urgency.

disagreement-5 Is PIN-AUDIT-PWR1.md's D1-PWR1 disposition still current?

LANE A {'documents': ['PIN-AUDIT-PWR1.md S(D1-PWR1)'], 'position': 'D1 pins 3/4/5/6 are bare floats, and the floating VBUS clamp node is the P0'}

LANE B {'documents': ['this consolidation, P-1 differential'], 'position': "those four pins LEFT the 14:58 floating list, and the only recorded mechanism in that window is Captain's NC sweep"}

RESOLUTION OPEN, and it is the reason RQ-005 exists. If the pins are NC-flagged the finding's severity RISES rather than falls: the sheet now certifies that a TVS clamping nothing is correct. P-1 settles it in one read.

disagreement-6 Did U3-PWR2.5 regress between the two DRC runs?

LANE A {'documents': ['PIN-AUDIT-MOTION.md S6'], 'position': "'one pin went the wrong way - U3-PWR2.5 became floating between the two runs', handed to A2-RAILS as a regression"}

LANE B {'documents': ['PIN-AUDIT-PWR2.md PWR2-001', 'RECONCILE-RECEIPT-2026-08-28b.md'], 'position': 'it was never connected - the BUCK_PG wire has been off-sheet since it was drawn, and the 12:17 DRC ran BEFORE the PG pull-up transaction executed at ~04:56Z'}

RESOLUTION Effectively resolved in favour of lane B - the 12:17 log predates the transaction, so its silence is not evidence of a working connection. Recorded because the DRC single-network warning is NOT a reliable dangling-net detector: MOTION_INT_S3 was a one-pin net in the frozen source and appeared in neither run's list. Its silence is not evidence.

Ruled out — do not re-raise

10 findings were raised and then killed by evidence, including two of the orchestrator's own. Recorded so they are not rediscovered and re-queued.

withdrawn-1

NFC RFO2/RFI2 floating is a P0 wiring defect - 'only half the RF front end is represented'.

RAISED BY PIN-AUDIT-NFC.md NFC-D4, first revision

KILLED BY AN5592 Rev 1 S2.2 Figure 3 shows the single-ended antenna stage containing ONLY RFO1 and RFI1 - RFO2 and RFI2 do not appear at all: no termination, no tie, no dummy network. ST

support states it directly: 'only RFO1 / RFI1 is connected to the antenna and will be driven' (Travis Palmer, ST Employee, 2024-01-22). The captured chain maps onto AN5592 Figure 3 term for term: Lemc0=L2, Cemc0=C59, RQ_s=R42, Cs1=C60, Cp1=C61.

WHAT SURVIVES The missing DECISION (DEC-04), the unrecorded firmware register (single=1, rfo2=0), and the uncoded read-range trade. All real, all P1.

withdrawn-2 NFC-D2: NFC_IRQ reaches no host processor.

RAISED BY PIN-AUDIT-NFC.md, four successive revisions

KILLED BY The 14:58:52 DRC. U9-ESP has ZERO floating pins (down from 19), and the only two single-pin networks on the entire board are \$1N153914 and BUCK_PG - neither NFC-related. All 41 U9-ESP pins sit on multi-pin nets, and the name-to-net map puts pin 4 (IO4) on NFC_IRQ.

WHAT SURVIVES

withdrawn-3

The ESP32-S3 is electrically disconnected - U9-ESP has 0 of 41 pins connected.

RAISED BY USB-TOPOLOGY-AUDIT.md S3, PIN-AUDIT-PWR2.md PWR2-012, RECONCILE-RECEIPT first pass

KILLED BY The 14:58:52 DRC: U9-ESP has zero floating pins. Refuted earlier and independently too: Captain's 12:17 DRC ran 46 SECONDS AFTER the ACCEPTED repair that left the displacement in place, and reported 19 floating pins there, not 41 - so EasyEDA considered the module connected WITH the displacement present. EasyEDA's netlist does not require exact endpoint coincidence.

WHAT SURVIVES A DRAWING defect (RQ-057), real against SINGLE-SHEET-CONTRACT.md. The reconcile receipt's own correction says it: 'geometry is not automatically the electrical truth either.'

withdrawn-4 AUDIO_BCLK_RT is a stub to nowhere.

RAISED BY PIN-AUDIT-AUDIO.md, first read (withdrawn by the same lane)

KILLED BY Its far stub terminates at (2415, 4660) - inside the U6-RTC symbol extent but OUTSIDE the 98-ball readback, so the ball cannot be named. EasyEDA's DRC lists seven single-pin nets and AUDIO_BCLK_RT is NOT among them, so the connection exists.

WHAT SURVIVES UNVERIFIED as to which ball. Not a defect. 25 further stubs sit in the same coordinate band and need the full 196-ball readback.

withdrawn-5 **The RT1062 has 111 unconnected balls.**

RAISED BY 12:17 DRC as read by the orchestrator

KILLED BY The 14:58 DRC: the real number is 80. The 111 was inflated by the same mis-registration that produced withdrawn-3.

WHAT SURVIVES 80 balls are still unadjudicated in this queue - PIN-AUDIT-RT.md did not land.

withdrawn-6

NFC-D15: the two-island NFC_IRQ geometry is 'electrically correct', and the visible-wiring requirement is power-tree-only.

RAISED BY PIN-AUDIT-NFC.md NFC-D15 (superseded by the same lane, kept on the record)

KILLED BY The 'electrically correct' half was never established, and the citation was an under-read: schematic/SINGLE-SHEET-CONTRACT.md carries a GENERAL wiring doctrine, not a power-tree-only one.

WHAT SURVIVES

withdrawn-7

The LED-branch eFuse current limit is UNDETERMINED because R8-PWR2 shows a 10 kOhm supplier field.

RAISED BY PIN-AUDIT-LED.md S5 and PIN-AUDIT-MOTION.md S4.4, both first revisions, both withdrawn by their own lanes

KILLED BY supplierId is a stale library-inherited string on several parts; the orderable identity lives in Supplier Part, and R8's is C185418 / RC0402FR-073K48L - a genuine 3.48 kOhm giving a determined 0.96 A. That distinction is also the clean test for the seven fake DNPs: real-valued parts carry an LCSC Cxxxxxxx in Supplier Part, while the fake DNPs carry the inherited 10 kOhm string in BOTH fields. Verified by re-reading bom_flat.csv field by field.

WHAT SURVIVES

withdrawn-8

The buck feedback divider R5-PWR2 / R6-PWR2 is a third instance of the shared-10 kOhm mis-bind.

RAISED BY team-lead hypothesis, tested by PIN-AUDIT-PWR1.md

KILLED BY POSITIVE VERIFICATION, not absence. R5 sits on the all-100 k device with seven siblings, all 100 k; R6 (32.4 k) has a device of its own. Every other programming resistor, divider and timing component in the power-entry blocks was checked the same way and verifies clean.

WHAT SURVIVES The mis-bind is exactly two parts - R1-PWR1 and R8-PWR2 - and the fix is bounded. R6's VALUE is still wrong for a different, datasheet reason: RQ-024.

withdrawn-9-minor Add into BOM does not exist in the schematic source.

RAISED BY CANVAS-OWNERSHIP-PROBE first pass

KILLED BY It exists on four records (C43-ESP, C44-ESP, C52-AUD, U6-RTC part 2) and ranks 44th of 213 attribute keys, below the cut-off of the truncated list it was read from. index.json's own attr_key_census records 'Add into BOM': 4.

WHAT SURVIVES This makes the seven DNP resistors a MEASURED defect rather than an inference.

withdrawn-10-minor

sources/SOURCE-REGISTER.md registers only AN5240, and SLUSEA4 is unregistered.

RAISED BY brief

KILLED BY The register already carries three ST rows: the ST25R3916B datasheet, the STEVAL-ST25R3916B reference design, and AN5240. And SLUSEA4 IS registered - but it is the TPS62933 datasheet, a different converter in a different package with a different pinout; the TPS62913 document self-identifies as SLVSFP4B.

WHAT SURVIVES D-045's two substantive claims were independently re-derived from SLVSFP4B, so D-045 STANDS ON ITS MERITS - only the citation is wrong. Six documents relied on across the audits are unregistered: SLVSFP4B, SBVS050M, SBVS320D, SLVSFC9C, SCLS378P, LCSC C185418.

Generated from repair-queue.json · frozen denominator 489736:464c27d4 · netlist evidence from the 14:58:52 GUI DRC · live has moved past the frozen hash, so every entry is a proposal for the single writer to reconfirm.